MFC Integration — Simple Steps

1. Copy these into the MFC project

- `Stitcher.dll`
- `Stitcher.lib`
- `stitcher_api.h`
- `microstitch/`

2. Link Stitcher.lib in the MFC project

3. Include the header

```
#include "stitcher_api.h"
```

4. Create the stitcher once when the camera starts

```
void* stitcher = Stitcher_Create(640, 360, 3);
```

5. Get the USB microscope frame as `cv::Mat frame`

6. Show `frame` in MFC Window 1 → Live Camera

7. Send the same frame to the DLL

```
Stitcher_ProcessFrame(
    stitcher,
    frame.data,
    frame.cols,
    frame.rows,
    frame.channels(),
    (int)frame.step
);
```

8. Get the updated mosaic

```
const unsigned char* data;
int width, height, channels, stride;

Stitcher_GetMosaic(
    stitcher, &data, &width, &height, &channels, &stride
);
```

9. Convert the returned data to `cv::Mat`

```
cv::Mat mosaic(
    height, width, CV_8UC3,
    const_cast<unsigned char*>(data),
    stride
);
```

10. Show `mosaic` in MFC Window 2 → Live Mosaic

11. When closing the camera/app

```
Stitcher_Destroy(stitcher);
```

Final flow

```
USB Microscope
      |
```

```
      v
cv::Mat frame
      |--------> Window 1: Live Camera
      v
Stitcher.dll
      |
      v
cv::Mat mosaic
      |
      v
Window 2: Live Mosaic
```

That's essentially all he needs to connect your backend to his existing MFC application.